MASY1-GC 1800 | ASSIGNMENT 6

Primary Agent Output

Robotic-assisted total knee replacement | hypothetical teaching hospital A | v1.1

Recommendation: conditional bounded experiment

Plan a comparative pilot under the stated teaching-hospital capability assumptions. Do not enrol patients until safety, permissions, training and protocol gates are verified. If those conditions cannot be established, defer. Do not implement routinely now.

Technology generally

Technical maturity: a clinically used surgeon-controlled platform, not an autonomous prototype. Diffusion demonstrates use, not superior patient value. Comparative evidence does not establish a consistent recovery advantage. General evidence is identical across these hypothetical hospitals (S1-S4).

Application finding

For primary elective total knee replacement, the relevant benefit is earlier walking and return to daily activities, not alignment accuracy alone. Conventional knee replacement with rehabilitation remains the comparator. Same eligibility and rehabilitation should be retained where appropriate; partial-knee and other-platform findings are not direct proof for this application.

Organization-specific finding

Hospital A is assumed to have the people and research resources to resolve a useful local implementation question. This supports planning controlled learning, not claiming readiness already verified. A research-oriented posture tolerates uncertainty about incremental value, not an unassessed safety risk.

Decisive judgment

The hospital's assumed research capacity supports disciplined learning. Its size or enthusiasm does not establish safety. The student's value criterion is faster walking and daily-activity recovery, with safety preserved and any additional cost justified.

Alternatives, uncertainty and reversibility

Retain conventional surgery and assess rehabilitation/workflow improvements. A pilot budget can be capped; injury cannot be reversed by cancelling the programme. Avoid sunk-cost, novelty and peer-pressure reasoning. Also avoid rejecting all experiments solely from risk aversion.

Pilot conditions and contrast

Management-level proposal; clinical and statistical protocol design remains with qualified reviewers.

What the pilot should measure

Use a defined conventional-surgery comparator and comparable eligibility/rehabilitation. Measure days to prespecified walking and daily-activity milestones, supported by functional assessments. Track complications, readmissions, malfunctions, conversions and full costs. Baseline and 2-, 6- and 12-week assessments are proposed planning intervals, not published requirements.

Safety and decision rules

An orthopaedic clinical lead and independent safety oversight should own the evaluation. Pause new enrolment after an unexpected serious complication or repeated equipment failures, while patient care continues. Investigate cause and require authorized corrective action before restarting. Clinical reviewers must define thresholds, meaningful recovery differences, safety margins, allocation and sample size before launch.

When the recommendation changes

Review within 90 days or earlier for safety alerts, new evidence or readiness changes. Resolve the trial correction and verify platform authorization. Expansion requires useful recovery improvement, credible safety evidence and an acceptable local total-cost case; a small pilot with no adverse events cannot prove rare-harm equivalence.

Contrast and stress-test results

Hospital B: DEFER because training, support and outcomes capacity are missing. Aggressive hospital A without completed training/safety review: DEFER as well. Both retain exactly the same general technology and application findings. If A's capabilities also prove absent in practice, its conditional pilot recommendation becomes deferral.

Confidence and boundaries

Moderate confidence in the management posture, low confidence in a local recovery advantage. No real hospital audit or contract was supplied. The Organizational Adoption Agent must verify readiness; clinicians retain patient selection and safety authority.

Evidence and verification

Sources retrieved 6 November 2026. Publication dates and applicability limits are retained below.

S1 | Official health technology guidance

NICE permits specified systems under evidence-generation conditions, rather than endorsing unrestricted routine adoption. Staff training is required.

Published 2025-04-17; retrieved 2026-11-06. England/NHS context. Search-indexed page text retrieved; direct fetch returned 403. Regulatory and local permissions must be verified for the specific system.

S1 | NICE HTG743, Recommendations | https://www.nice.org.uk/guidance/htg743/chapter/1-Recommendations

S2 | Official evidence assessment

NICE found more precise alignment but uncertain translation into better patient outcomes; evidence differs across systems and procedures.

Published 2025-04-17; retrieved 2026-11-06. Do not transfer results from partial replacement or another platform without qualification.

S2 | NICE HTG743, Committee discussion | https://www.nice.org.uk/guidance/htg743/chapter/3-Committee-discussion

S3 | Multicentre randomized trial abstract

The RACER-Knee randomized trial reported no meaningful 12-month patient benefit for Mako compared with conventional instruments and similar observed safety profiles.

Published 2026-09-05; retrieved 2026-11-06. Search-indexed abstract reviewed. A correction dated 2026-10-24 is indexed (doi:10.1016/S0140-6736(26)02109-4); correction text was not retrieved. Use qualitative conclusion provisionally, not a numerical procurement threshold.

S3 | Parsons et al., Lancet 2026;408:924-934; doi:10.1016/S0140-6736(26)00986-4 | https://pubmed.ncbi.nlm.nih.gov/42624811/

S4 | Original research institution report

The trial team reports comparable walking and early pain outcomes and higher resource use for robotic surgery in RACER-Knee.

Published 2026-08-21; retrieved 2026-11-06. Full page read. Corroborates S3 but is the same trial, not independent replication; predates indexed correction. Manufacturer supported additional robotic procedure costs.

S4 | University of Warwick, trial-team release | https://warwick.ac.uk/news/pressreleases/robotic-assisted-knee/

S5 | Randomized trial abstract

A 68-patient single-centre ROSA randomized trial reported some early pain and flexion benefits, with no reported adverse events in its small sample.

Online 2026-04-17; retrieved 2026-11-06. Different platform and setting; cannot establish Mako benefit or rule out rare harm. Pain/flexion are not direct proof of earlier independent daily activities.

S5 | Neira et al., Knee 2026;61:104457; doi:10.1016/j.knee.2026.104457 | https://pubmed.ncbi.nlm.nih.gov/42000386/

Access and interpretation limits

Some original-source text was available through search indexing; the direct NICE fetch was blocked and PubMed direct opens were incomplete. No full-text review is claimed. The correction to S3 was identified but not resolved. S4 is a same-study corroboration; S5 tests a different platform. These gaps prevent an unconditional clinical recommendation.